

Reading pack

Three core readings, two further readings, and a guide to using the figures

Before class

Read the abstracts and selected sections below. Focus on the argument and figures; detailed mathematical derivations are optional for this lecture. The three core readings move from model behavior to artistic intention to filmmaking practice.

A useful note-taking pattern

For each figure, record: the task; the input or condition; what changes; what stays stable; and what the example cannot establish. Keep a distinction between a selected demonstration and evidence of reliability.

After class

Choose one core reading for Quiz question 5. Use a specific section, figure or case rather than a general impression. The two further readings extend the lecture into representations, supporting tools and evaluation.

Survey of Video Diffusion Models: Foundations, Implementations, and Applications

Yimu Wang et al. | 2026 revision; TMLR

[Read the original paper](#)

Lecture slides 5-32 | Selected reading: Sections 1; 2.1 and 2.5; 4.1, 4.4 and 4.5. Skim Section 3.3.

How do conditions and temporal connections turn image generation into controllable video?

Read the map before the mathematics

Start with the introduction and video generative paradigms. Identify the difference between generating an entire sequence, generating progressively, and using an existing image as a condition. You do not need to derive the learning objectives for this lecture.

Follow information through the model

In the architecture figures, locate the visual input, text or other conditions, spatial processing, temporal processing, and output. Explain why a sequence needs relationships across frames. The diffusion time step describes noise removal; it is different from the time axis inside a video.

Choose a useful condition

Read Conditions, Consistency and Long video. For a shot of a painting moving inside a still room, identify a condition that specifies appearance and another that specifies movement or structure. Explain which parts remain unspecified.

Read evaluation critically

A selected sequence can illustrate a capability without establishing reliability. Ask how identity, temporal continuity, action correctness, camera motion and visual quality are measured. Longer output does not by itself establish story coherence.

Try it

Sketch a three-shot art video. For one shot, name the input condition, the property to preserve, a likely failure, and a visible test.

Check your understanding

Can you explain why four attractive individual frames may still make a poor video?

Diffusion-Based Visual Art Creation: A Survey and New Perspectives

Bingyuan Wang, Qifeng Chen and Zeyu Wang | ACM Computing Surveys, 2025; preprint 2024

[Read the original paper](#)

Lecture slides 33-50 | Selected reading: Sections 4.2, 5.1, 5.3, 5.4 and 6. Read the framework figures alongside the text.

How can an artistic intention become a technical intervention and an audience experience?

Begin with the artistic requirement

The survey connects artistic requirements and technical problems. Describe an intended experience in concrete terms: a delayed reveal, uneasy repetition, a sense of memory, or a shift in attention. A style label alone does not describe an entire artistic intention.

Translate intention into something controllable

Name a visible or temporal property that can support the intention: composition, material, color, subject identity, timing, or the relationship between foreground and background. Then identify an input or edit that can influence it.

Separate capability from interpretation

A method may change texture or preserve layout. Whether that change expresses nostalgia or unease depends on context and viewers. Explain the proposed connection, then use comparison or audience feedback to test it.

Read the discussion as an invitation to combine perspectives

The technical and synergistic discussions help connect tools to creative practice. Do not turn the survey into a ranking of generators. Ask what decisions remain with the creator and how the workflow makes those decisions possible.

Try it

Make two versions of the same shot plan. Change only one property and predict how the audience experience will differ. Ask a peer what they actually notice.

Check your understanding

Can you distinguish an artistic goal from the technical operation used to pursue it?

Generative AI for Film Creation: A Survey of Recent Advances

Ruihan Zhang et al. | CVPR 2025 CUEU workshop

[Read the original paper](#)

Lecture slides 51-72 | Selected reading: Sections 2.2 and 3; then choose two cases from Sections 4.1-4.6.

How do creators turn generated material into usable shots and a finished sequence?

Establish what the evidence describes

The Film Hack samples contain 8, 67 and 118 films in 2023, 2024 and 2025. They describe selected participants in a particular setting, not industry-wide adoption. The 2025 mean of 3.14 video tools per film is a sample statistic, not a recommended recipe.

Read the ratings correctly

The importance/performance comparison uses 100 respondents and a 0-7 scale. It concerns artists' reported assessments of character identity, body movement, camera control and local editing. It is not an objective benchmark of current products.

Trace two contrasting workflows

Compare, for example, Overthinking and For Pixi, or Clown and Fish Tank. Locate source material, generation or adaptation, manual correction, animation, and editing. Explain which stage contributes a visible feature in the final work.

Treat continuity as a considered choice

A changing appearance can support fragmented identity, while a recurring character may require stable features. Ask what must remain legible for the audience and which variation serves the work. Selected stills cannot establish the quality of the full moving sequence.

Try it

Draw a three-shot plan for the studio scene. Assign a reference, generation step, revision step and evaluation check to each shot. Keep only tools that solve an identified problem.

Check your understanding

Can you explain one case without attributing every effect to the video generator?

Further reading

State of the Art on Diffusion Models for Visual Computing

Ryan Po et al. | Computer Graphics Forum, 2024

[Read the original paper](#)

Slides 73-87 | Sections 3.3-3.4, 5.2, 6.2 and 7.

Compare pixels, geometry, scene representations and movement data. Ask which later edits each representation supports.

Advances in Artificial Intelligence: A Review for the Creative Industries

Nantheera Anantrasirichai, Fan Zhang and David Bull | Artificial Intelligence Review, 2026

[Read the original paper](#)

Slides 88-100 | Sections 3.1, 3.3-3.5, 3.7 and 4.

Place generation alongside enhancement, segmentation, tracking, spatial capture and evaluation. Historical examples are not a current product ranking.